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ENGL393

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Technical Instructions

The context for my technical instructions is Terrapin Works, a network of digital manufacturing and prototyping facilities operated by the A. James Clark School of Engineering at the University of Maryland. Terrapin Works manages multiple labs that support research and instruction through access to over 200 3D printers. In this setting, I am a student employee in the Rapid Prototyping Center (RPC), responsible for preparing and maintaining consumer-grade printers such as the Prusa i3 MK3S+ for research and instructional use. My intended audience includes new undergraduate research assistants and student employees in the RPC who often have strong design software skills but limited hands-on experience with 3D printing hardware. They value precision, efficiency, and adherence to safety and workflow standards because these directly affect productivity and machine reliability. These instructions will help new assistants properly prepare, calibrate, and test 3D printers, reducing setup errors, conserving materials, and maintaining consistent lab operations.

Preparing and Calibrating the Prusa i3 MK3S+ 3D Printer at Terrapin Works (Rapid Prototyping Center)

Audience: Undergraduate student employees and research assistants at the University of Maryland's Terrapin Works Rapid Prototyping Center (RPC).

Purpose: To correctly prepare, calibrate, and verify the Prusa i3 MK3S+ 3D printer for safe, accurate printing.

Introduction

These instructions are written for new undergraduate research assistants and student employees at the University of Maryland's Terrapin Works Rapid Prototyping Center (RPC). The Prusa i3 MK3S+ is one of the most frequently used printers in the lab, and proper calibration is essential to ensure safety, accuracy, and material efficiency.

While many student operators have prior experience with design software like Fusion 360 or SolidWorks, few have hands-on knowledge of machine calibration. By following these steps, users will be able to independently prepare and calibrate a printer for production while maintaining the safety and procedural consistency expected at Terrapin Works (Terrapin Works, 2025; Greif & Mills, 2010).

Successful completion ensures:

- Correct mechanical alignment and extrusion
- Reduced print failures and wasted filament
- Compliance with lab safety and policy standards

Warnings / Cautions

- Thermal Hazards: The nozzle and bed reach over 200°C during calibration. Avoid contact with the print head and build surface (Prusa Research, 2022).
- Pinch Risk: Keep hands and loose clothing clear of moving parts
- Surface Damage Warning: Incorrect Z-offset calibration can scratch or gouge the PEI build sheet (Prusa Research, 2025).
- Filament Clogging: Never mix filament types without a full purge cycle. Contamination can cause jams and inaccurate calibration (Prusa Research, 2025).

- Policy Compliance: Only approved filament and software may be used on Terrapin Works printers (Terrapin Works, 2025). Non-compliance can result in access suspension.

Technical Background

1. Prusa i3 MK3S+ Key Systems

The Prusa i3 MK3S+ uses Fused Deposition Modeling (FDM) technology, where a thermoplastic filament is heated and extruded through a nozzle to build an object layer-by-layer. For successful printing, the first layer must be perfectly adhered to the build plate. The printer uses several automated systems to achieve this, but they require proper initial setup.

- **SuperPINDA Probe:** This is an inductive sensor on the print head that automatically detects the distance to the metal build plate. It is used to create a mesh of the bed's surface, accounting for any minor warps or tilts.
- **Mesh Bed Leveling (MBL):** Before a print, the printer probes the bed at multiple points to create a "mesh" or height map. During printing, it uses this map to make micro-adjustments to the print head height, ensuring the nozzle distance is consistent across the entire bed.
- **Live Adjust Z:** This is your most critical manual adjustment. While the MBL handles unevenness, the Live Adjust Z value sets the *overall* distance between the nozzle and the bed. An incorrect value is the most common cause of print failure:
    - **Too High:** The filament won't stick to the bed, resulting in a messy, spaghetti-like print.
    - **Too Low:** The nozzle will scrape the bed, obstructing filament flow and potentially damaging the build surface.

2. Required Prerequisites and Knowledge

Before you begin the physical calibration, you must have the following:

- **FDM Fundamentals:** You should understand basic 3D printing concepts, including:
    - The role of a **slicer** (software that converts a 3D model into printer instructions/G-code).
    - The importance of the **first layer** for bed adhesion.
    - What a **G-code** file is and how it is sent to the printer.

- **Software Proficiency:**
    - **Cura:** You must know how to open a 3D model (.stl file), select the correct printer profile, adjust basic settings (like infill and layer height), and slice the model to generate a .gcode file.
    - **PaperCut MF:** You must have an active account and know how to log in, upload a .gcode file to the print queue, and release the job to the printer at the machine itself.
- **Institutional Training:**
    - You **must** have completed the **Terrapin Works Safety Orientation**.
    - You **must** have completed the **RPC (Registered Print Coordinator) Policy Training** and be an authorized user.

Materials and Equipment

Category	Item / Software	Notes
Printer	Original Prusa i3 MK3S+	Assigned printer at RPC
Filament	1.75mm PLA	From Terrapin Works-approved storage
Software	Cura (latest RPC version)	For slicing calibration models
Computer Access	RPC workstation w/ PaperCut	For queue submission and print tracking
Tools	Lint-free cloth, isopropyl alcohol (70%+), tweezers, scraper, nozzle cleaning needle	For maintenance
Safety	Protective eyewear	During bed/nozzle heating

Calibration Files	Calibration cube or zig-zag test print	Provided in RPC digital library

Procedure: Calibrating the Prusa i3 MK3S+

Objective: To prepare the printer for a production job by ensuring it is clean, mechanically calibrated, and has a perfectly tuned first layer.

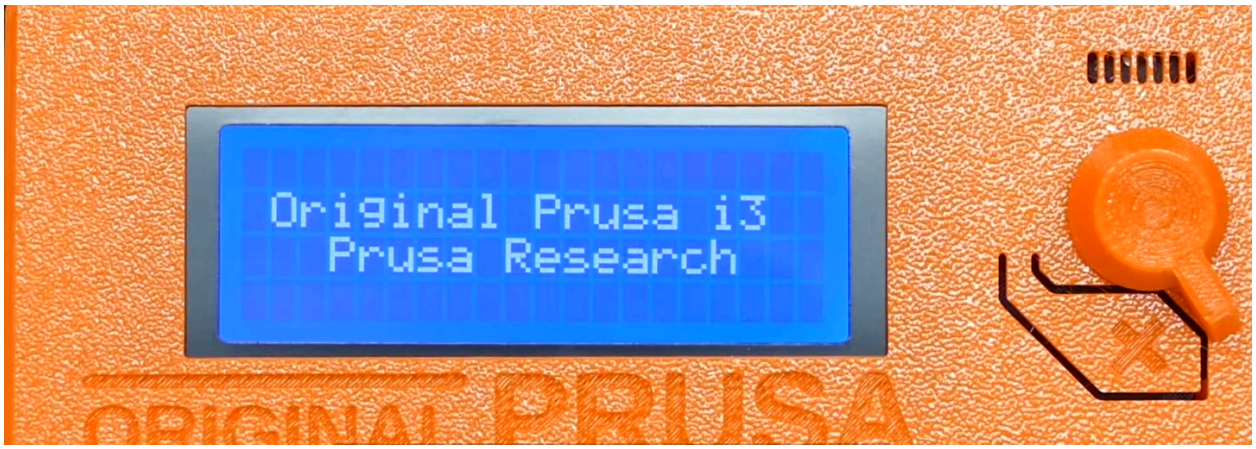
Step 1. Power On the Printer

Action:

1. Locate the power strip that the printer is plugged into.
2. Press the **ON** switch for the specific outlet controlling the Prusa i3 MK3S+.

Explanation: The printer will boot up and automatically run a self-test, checking its motors, sensors, and heating systems. Do not interact with the printer until this is complete

Result: The LCD screen will light up, display **"Original Prusa i3 Prusa Research"**, and then show a "Ready" status with no error messages.



Step 2. Inspect and Load Filament

Action:

- Confirm that the correct filament type (PLA) is installed.
- Use Unload Filament if another material is present.
- Select AutoLoad Filament and insert PLA until it extrudes cleanly.

Warning: Nozzle reaches 200°C; do not touch metal parts (Prusa Research, 2022). Result: Steady filament flow with no gaps or discoloration.



Step 3. Clean the Build Plate

Action:

- Remove the steel sheet from the bed.
- Wipe with isopropyl alcohol using a lint-free cloth.

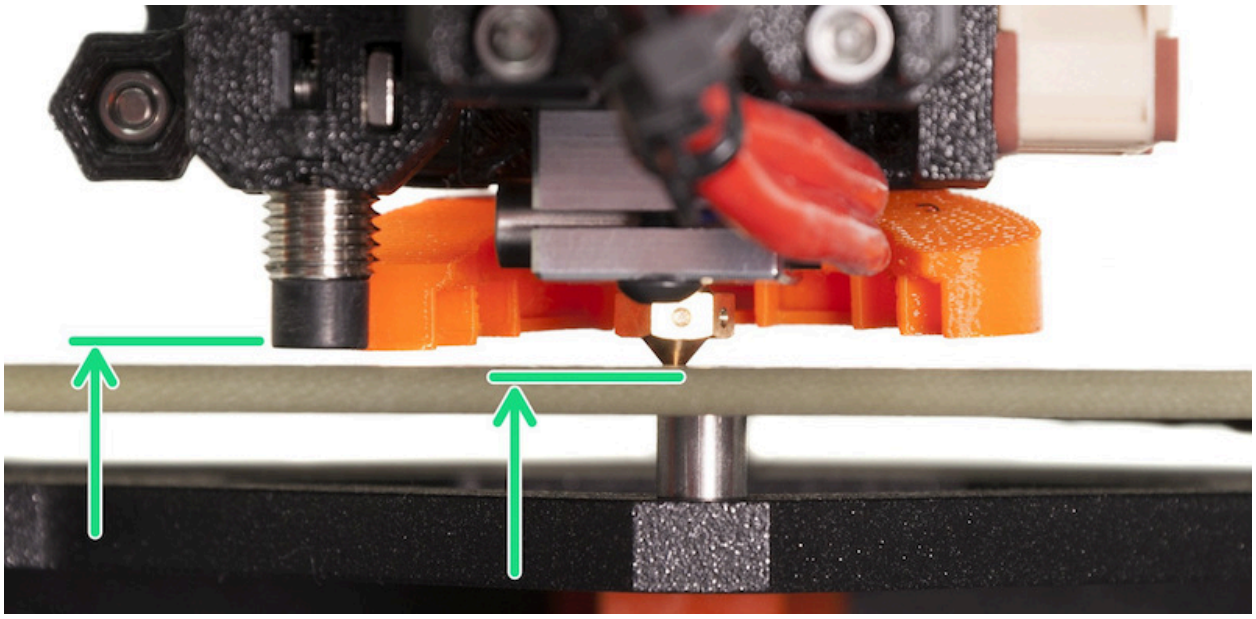
Explanation: Residue can prevent adhesion during first-layer calibration. Result: Smooth, dry surface free of oils or debris.

Step 4. Run XYZ Calibration

Action:

- On the LCD: navigate to Calibration → XYZ Calibration.
- Follow prompts to allow the printer to home all axes.

Explanation: This ensures the X, Y, and Z motion systems are square and aligned (Prusa Research, 2025b). Result: Screen displays “XYZ Calibration Successful.”



Step 5. Secure the Build Sheet

Action:

- Reattach the build sheet magnetically to the bed.
- Verify it sits flush without lifting edges.

Note: Switching between smooth and textured sheets requires new first-layer calibration (Prusa Research, 2025a). Result: Build sheet flat and secure.

Step 6. Perform Mesh Bed Leveling

Action:

- From LCD, select Calibration → Mesh Bed Leveling.
- Wait for the printer to heat and probe several points.

Explanation: The system maps height differences for consistent nozzle spacing (Prusa Research, 2025a). Result: “Mesh Bed Leveling Complete” displayed.

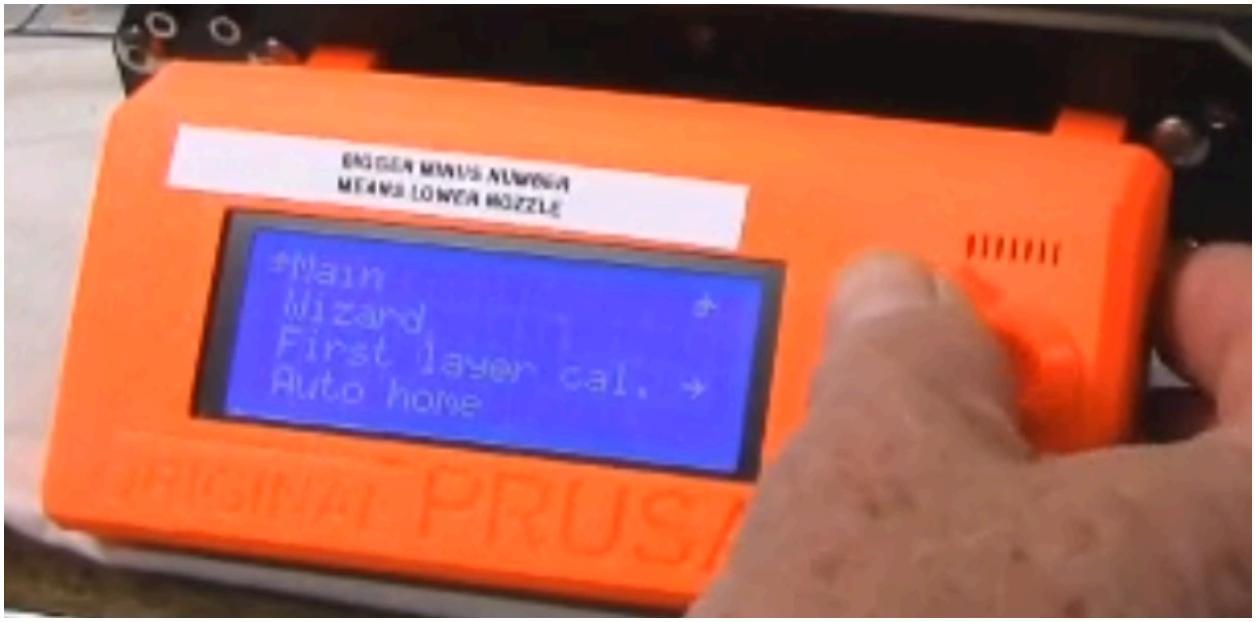
Step 7. Conduct First Layer Calibration

Action:

- Navigate to Calibration → First Layer Calibration.

- Adjust the Z-offset during the test pattern until the filament appears slightly squished and smooth.

Warning: Too low → nozzle scratches bed; too high → filament won't stick. Result: Continuous, glossy lines adhering evenly (Prusa Research, 2025, February).



Step 8. Slice and Prepare Calibration Model

Action:

- Open Cura on the RPC workstation.
- Load the calibration cube file from the RPC library.
- Confirm printer profile = *Prusa i3 MK3S+*, material = PLA.
- Export G-code to SD card or submit via PaperCut system (PaperCut, 2025).

Explanation: This verifies slicing accuracy and job tracking. Result: Calibration cube G-code ready for printing.

Step 9. Begin Calibration Print

Action:

- Insert SD card or queue job from PaperCut.
- Start print and monitor the first 2–3 layers.

Warning: Do not leave the printer unattended during first-layer printing. Result: Smooth, consistent layers with proper adhesion.

Step 10. Post-Print Procedure

Action:

- Allow the bed to cool below 35°C.
- Gently remove the print with a scraper.
- Wipe the bed again and return all tools.

Explanation: Cleaning prevents residue buildup and ensures reliable future adhesion. Result: Printer reset and ready for next user.

Step 11. Record and Report

Action:

- Log completion in the RPC print record or queue.
- Report persistent calibration issues to your graduate mentor.

Explanation: Consistent logging enables team coordination and maintenance tracking (Greif & Mills, 2010). Result: Documentation complete and machine status updated.

Conclusion

By completing these procedures, student operators ensure their assigned printers are clean, calibrated, and compliant with Terrapin Works safety and quality standards. Following both manufacturer guidelines (Prusa Research) and Terrapin Works policies reduces machine downtime and ensures accuracy in academic research.

References

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Reflection

I edited the document to make the terminology less dense, especially in the explanatory sections before the procedural steps, so the reader could follow along more easily. Instead of the peer review, I worked with an instructor at the Writing Center. The instructor emphasized that while my content was technically accurate, the tone was overly formal and sometimes detached. They suggested making the writing more readable for new lab assistants by replacing heavy jargon with simpler phrasing and by integrating the purpose of the document more naturally into the introduction. For example, instead of explicitly stating, "My intended audience is undergraduate research assistants," I revised it to flow naturally within the context of the lab and the procedures at Terrapin Works. This made the instructions sound more authentic and reader-centered.

My instructor's feedback focused on structure and clarity. They mentioned that while my steps were well-organized and detailed, my original version lacked visual hierarchy and consistent formatting for the procedure sections. They recommended using bolding and bullet points to clearly distinguish between actions, explanations, and results. I addressed this by standardizing each step to include "Action," "Explanation," "Result," and "Warning/Note" subsections. This not only improved readability but also aligned my document more closely with professional technical writing conventions.

For resources, I used the 7.7 Writing Instructions section from the Learning Module. I used its detailed breakdown of how to structure instructions like grouping tasks logically, providing clear introductions, using numbered lists, and emphasizing readability through formatting to shape my procedural layout. The module also stressed the importance of understanding the audience's familiarity with the topic, which guided how I wrote my background section and simplified the language in my steps.

From my references, I used Greif and Mills (2010), who discuss the process of managing undergraduate research assistants. Their discussion of setting up lab procedures and maintaining consistency across team members helped me understand how clear, repeatable instructions contribute to research efficiency and training quality. The excerpt on "Choosing a laboratory structure" and "Formulating goals" inspired me to think about the long-term purpose of my instructions which is training new assistants to work independently while maintaining lab standards. Prusa Research's technical manuals also provided step-by-step details that ensured my instructions were accurate and aligned with manufacturer procedures.

Over the course of this assignment, I learned that my writing tends to default to overly technical language, especially when explaining processes I am already familiar with. By focusing on accessibility and user perspective, I realized that clarity and structure are just as important as accuracy in technical writing. This project taught me how to balance detail with simplicity and how to use formatting strategically to make information more navigable. In future projects, I plan to apply these lessons by drafting with the reader's experience in mind first and refining the technical precision afterward.